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1 Basics

1.1 单位分解

Theorem 1.1.1 ▶ 单位分解的存在性

设 M 是一个流形, $\mathscr{U} = \{U_\alpha : \alpha \in A\}$ 是 M 的一个开覆盖, 则

1. 存在从属于 $\mathscr{U}$ 的单位分解 $\Phi = \{\varphi_i : i \in I\}$ 且 I 是至多可数的, $\operatorname{supp}\varphi_i$ 是紧的.
2. 存在从属于 $\mathscr{U}$ 的具有相同指标集的单位分解 $\Phi = \{\varphi_\alpha : \alpha \in A\}$, 且至多可数多个 $\varphi_\alpha \neq 0$.
3. 如果 $\mathscr{U}$ 局部有限, 且 $\overline{U_\alpha}$ 是紧的. 则存在从属于 $\mathscr{U}$ 的具有相同指标集的单位分解 $\Phi = \{\phi_\alpha : \alpha \in A\}$ 且 $\operatorname{supp}\varphi_\alpha$ 是紧的. [a]

[a] 由于紧急的闭子集是紧的, 所以第三条可以有第二条简单地导出.

Corollary 1.1.2

设 M 是一个流形, $D \subseteq M$ 是闭集, $U \subseteq M$ 是 D 的一个开邻域. 则存在 $f \in C^\infty(M)$, 使得 $f|_D \equiv 1$, 且 $\operatorname{supp} f \subseteq U$.

Proof. 考虑 M 的一个特殊的开覆盖, $\{U, M \setminus D\}$. 对它应用上面定理的第二种情况即可. □

1.2 向量场和微分算子

Definition 1.2.1

1. 设 M 是一个流形, $p \in M$. 若 $f, g \in C^\infty(M)$ 满足, 存在 p 的开邻域 U, 使得 $f|_U = g|_U$, 则称 f 和 g 在 p 附近等价.
2. 称 $C^\infty(M)/\sim$ 为 p 附近的函数芽, 记作 $C_p^\infty(U)$.
3. 对于 $f \in C_p^\infty(M)$, 它的芽记作 $[f] \in C_p^\infty(M)$

在欧式空间上, 光滑函数有展开式

$$f(x)=f(x_0)+\sum_{i=1}^{m}\frac{\partial f}{\partial x^i}(x-x_0)^i+\sum_{i,j=1}^{m}\frac{1}{2}\frac{\partial^2 f}{\partial x^i\partial x^j}(x_0)(x-x_0)^i(x-x_0)^j+O\left(\|x-x_0\|^2\right)$$

$$D=C(x)+\sum_{i=1}^{m}a_i(x)\frac{\partial}{\partial x^i}+\sum_{i,j=1}^{m}\frac{1}{2}b_{ij}(x)\frac{\partial^2}{\partial x^i\partial x^j}$$

Definition 1.2.2

令 $m_p=\left\{f\in C_p^\infty(M):f(p)=0\right\}$, 则 m_p 是 $C_p^\infty(M)$ 的一个极大理想. 特别地, m_p^k 也是 $C_p^\infty(M)$ 的理想.

则 f 的展开式的第二项和第三项分别视作 m_p 和 m_p^2 中的元素.

Definition 1.2.3 ▸ Differential Expression of Order r

1. 一个 r 阶微分表达式, 是指一个线性函数 $D:C_p^\infty(M)\to\mathbb{R}$ 满足 $D\left(m_p^{r+1}\right)=0$.
2. 全体 p 处的 r 阶微分表达式记作 $T_p^{(r)}(M)$
3. 定义 $T_p^{(\infty)}(M)=\bigcup_{r\geq 0}T_p^{(r)}(M)$

在这个定义下, 当 $r=2$ 时,

$$\begin{aligned}D(f(x))&=D\left(f(x_0)+\sum\frac{\partial f}{\partial x^i}(x-x_0)^i+\sum\frac{1}{2}\frac{\partial^2 f}{\partial x^i\partial x^j}(x-x_0)^i(x-x_0)^j\right)\\&=D(f(x_0))+\sum_i\frac{\partial f}{\partial x^i}(x_0)D\left((x-x_0)^i\right)+\sum_{i,j}\frac{1}{2}\frac{\partial^2 f}{\partial x^i\partial x^j}(x_0)D\left((x-x_0)^i(x-x_0\right.\\&=:cf(x_0)+\sum_i a_i\frac{\partial f}{\partial x^i}(x_0)+\sum_{i,j}\frac{1}{2}b_{ij}\frac{\partial^2 f}{\partial x^i\partial x^j}(x_0)\end{aligned}$$

其中 $c,a_i,b_{i,j}$ 均为常数.

Definition 1.2.4 ▶ 切向量

1. 对于一个 1 阶微分表达式 X, 若它满足

$$X(f\cdot g)=X(f)\cdot g(x_0)+f(x_0)X(g)$$

则称 X 为 p 处的一个切向量.
具体地,

$$X(f)=\sum_{i=1}^{m}a_i\frac{\partial f}{\partial x^i}(x_0)$$

2. 记 $T_p(M)$ 为 p 处全体切向量构成的线性空间. 则 $T_p(M)\simeq\mathbb{R}^m$. $\partial_i:=\frac{\partial}{\partial x^i}$ 构成 $T_p(M)$ 的基.

让 $p\in U$ 在 U 上可变 $\iff$ $x\in V$ 跑起来. 则 $X=\sum a^i(x)\frac{\partial}{\partial x^i}$, $a^i\in C^\infty(u)$.

Definition 1.2.5 ▶ 多重指标

- 一个多重指标 α 是指 $\alpha=(\alpha_1,\cdots,\alpha_m)$, $\alpha_i\in\mathbb{N}$. 定义 α 的长度为 $|\alpha|=\alpha_1+\cdots+\alpha_m$
- 定义记号

$$\partial^\alpha:=\left(\frac{\partial}{\partial x^1}\right)^{\alpha_1}\cdots\left(\frac{\partial}{\partial x^m}\right)^{\alpha_m}$$

Proposition 1.2.6

$T_p^{(r)}(M)$ 的一组基为 $\{\partial^\alpha:|\alpha|\le r\}$

Definition 1.2.7

一个线性映射 $\mathcal{D}:C^\infty(M)\to C^\infty(M)$ 是一个阶数至多为 r 的微分算子，如果对于流形 M 上的任意一个局部坐标图 $(U,x^1,\dots,x^m)$，算子在 U 上的限制 $\mathcal{D}|_U$ 都可以表示为:

$$\mathcal{D}|_U=\sum_{|\alpha|\le r}a_\alpha(x)\partial^\alpha$$

其中:

- $\alpha=(\alpha_1,\dots,\alpha_m)$ 是多重指标.
- $\partial^\alpha=\frac{\partial^{|\alpha|}}{(\partial x^1)^{\alpha_1}\cdots(\partial x^m)^{\alpha_m}}$ 是偏微分.

- 每一个系数 $a_\alpha(x)$ 都是定义在 U 上的光滑函数，即 $a_\alpha \in C^\infty(U)$.

Definition 1.2.8

- 开集 U 上的切向量的全体记作 $\mathfrak{X}(U)$.
- 开集 U 上的微分算子的全体记作 $D(U)$

Remark.
$D(U)$ 上按算子的复合, 可以定义出自然的乘法. 它是结合但不交换的.

Definition 1.2.9

$\mathfrak{X}(U)$ 上有 Lie 括号

$$\begin{aligned}[,] : \mathfrak{X}(U) \times \mathfrak{X}(U) &\to \mathfrak{X}(U) \\ [X, Y] &:= XY - YX\end{aligned}$$

Proposition 1.2.10

- $C^\infty(U)$ 是交换结合的 $\mathbb{R}$-代数.
- $\mathfrak{X}(U)$ 和 $D(U)$ 都是 $C^\infty(U)$ 模

Proposition 1.2.11

若 D_1 是 r_1 阶的微分算子, D_2 是 r_2 阶的微分算子, 则
- $D_1 \cdot D_2$ 的阶数 $\leq r_1 + r_2$.
- $D_1D_2 - D_2D_1$ 的阶数 $\leq r_1 + r_2 - 1$

Theorem 1.2.12

设 M 是一个流形, $\dim M = m$. $X_1, \cdots, X_m \in \mathfrak{X}(M)$ 满足对于任意的 $p \in M$. $X_1(p), X_2(p), \cdots, X_m(p)$ 构成 $T_p(M)$ 的一组基[a]. 记 $X^\alpha = X_1^{\alpha_1} \cdots X_m^{\alpha_m}$. 则
- $\forall p \in M$, $\left\{X^\alpha_{(p)} : |\alpha| \leq r\right\}$ 构成 $T_p^{(r)}(M)$ 的基.
- 等价地说, $\{X^\alpha : \alpha = (\alpha_1, \cdots, \alpha_m)\}$ 构成 $D(M)$ 的作为 $C^\infty(M)$ 模的基.
- 此时, D 可以写作

$$D = \sum a_\alpha X^\alpha, \quad a_\alpha \in C^\infty(M)$$

[a] 即 M 是一个可平行化的流形

1.3 积分

设 M 是流形, $\dim M = m$, ω 是 M 上的一个 m-形式. $f \in C^{\infty}(M)$.

1. 取 M 的一个图册 $\mathcal{U} = \{(U_\alpha, \varphi_\alpha) : \alpha \in A\}$.
2. 取从属于 $\mathcal{U}$ 的一个紧支单位分解. $\Phi = \{\varphi_i : i \in I\}$
3. 则 $f = \sum_{i \in I} f \cdot \varphi_i$. 记 $f_i = f\varphi_i$
4. 设 $\operatorname{supp} \varphi_i \subseteq U_\alpha$.
5. 设 V_α 中的坐标为 $\left(x^1, \cdots, x^n\right)$, $f_i : U_\alpha \to \mathbb{R}$. 定义 $\tilde{f}_i = f_i \circ \varphi_\alpha^{-1} : V_\alpha \to \mathbb{R}$.
6. 设 ω 在 U_α 上有表达式 $\omega = \omega_\alpha(x)\, \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^m$.

Definition 1.3.1 ▶ 对密度的积分

定义

$$J_i := \int_{U_\alpha} \tilde{f}_i(x) \left|\omega_\alpha(x)\right| \mathrm{d}x^1 \cdots \mathrm{d}x^n$$

定义

$$J := \int_M f\,|\omega| := \sum_{i \in I} J_i$$

称为是对密度的积分.

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Remark.
需要证明 J 与紧支单位分解的选取无关.

Definition 1.3.2 ▶ 可定向性

对于流形 M. 如果存在一个图册 $\{U_\alpha : \alpha \in A\}$ 使得任意两个坐标卡的之间的转移函数的 Jacobi 行列式都是正的. 则称 M 是可定向的.

- 称这样的图册为定向图册
- 若 $\mathcal{U}_1$ 和 $\mathcal{U}_2$ 是两个定向图册, 满足 $\mathcal{U}_1 \cup \mathcal{U}_2$ 还是定向图册, 则称 $\mathcal{U}_1$ 和 $\mathcal{U}_2$ 是定向相容的.

Definition 1.3.3 ▶ 对形式的积分

若 M 还是可定向的, 可以定义

$$J_i = \int_{U_\alpha} \tilde{f}_i(x)\,\omega_\alpha(x)\,\mathrm{d}x^1 \cdots \mathrm{d}x^n$$

定义

$$J := \int_M f\omega := \sum_{\in I} J_i$$

Remark.
可以证明, J 不依赖于与 $\mathcal{U}$ 定向相同的图册的选取.

Definition 1.3.4

M 是可定向的, 当且仅当 M 上存在一个处处非零的 m- 形式.

Theorem 1.3.5

设 M 是可定向的带边流形, ω 是一个紧支的 $(m-1)$ 形式, 则

$$\int_M \mathrm{d}\omega = \int_{\partial M} \omega$$

1.4 映射与子流形

Lemma 1.4.1

设 M, N 是 C^∞ 流形, $\varphi : M \to N$ 连续. 若对于任意的 N 的开集 V, 和 $g \in C^\infty(V)$. $g \circ \varphi \in C^\infty\left(\varphi^{-1}(V)\right)$. 则 φ 是 C^∞ 的.

Lemma 1.4.2

对于 $y \in N$ 附近的两个 C^∞ 函数 g_1, g_2, 若 $g_1 \sim_y g_2$(函数芽). 则 $g_1 \circ \varphi \sim_x g_2 \circ \varphi$.

Definition 1.4.3 ▶ 拉回映射

M,N,φ 同前. 则 φ 诱导出线性映射

$$\begin{aligned}\varphi^* : C_y^\infty(N) &\to C_x^\infty(M)\\ g &\mapsto g\circ\varphi\end{aligned}$$

称为是关于 φ 的拉回映射

Definition 1.4.4 ▶ 切映射

M,N,φ 同前. 则 φ 诱导出对偶空间（微分算子空间）之间的映射

$$(\mathrm{d}\varphi)_* : C_x^\infty(M)^* \to C_{\varphi(x)}^\infty(N)^*$$

通过做限制得到映射

1. $$(\mathrm{d}\varphi)_x : T_xM \to T_{\varphi(x)}N$$
2. $$(\mathrm{d}\varphi)_x^{(\infty)} : C_x^\infty(M)^* \to C_{\varphi(x)}^\infty(N)^*$$

Remark.
这里蕴含了一些事实

1. 如果 X 是切向量, 则 $(\mathrm{d}\varphi)_x(X)$ 也是.
2. $\varphi^{-1*}(m_y)\subseteq m_x$, 从而 $\varphi^{-1*}(m_y^k)\subseteq m_x^k$.
3. 若 D 是一个 r 阶微分表达式, $(\mathrm{d}\varphi)_x(D)$ 也是 r 阶微分表达式.

Definition 1.4.5

M,N,φ 同前. 若 X 是 M 上的向量场. 若 $X\in\mathfrak{X}(M), Y\in\mathfrak{X}(N)$ 满足

$$\forall_{x\in M}(\mathrm{d}\varphi)_x(X(x)) = Y(\varphi(x))$$

则称 X 和 Y 是 φ -相关的.

Lemma 1.4.6

设 D 是 M 上的微分算子. $D : C^\infty(M)\to C^\infty(M)$. 若 $\tilde{D}\in D(C^\infty(N))$. 则 $\tilde{D} : C^\infty(N)\to C^\infty(N)$

Definition 1.4.7

$\varphi : M \to N$ 诱导出映射

$$\varphi^* : \Omega^*(N) \to \Omega^*(M)$$

Definition 1.4.8

设 M, N, φ 同前. $\dim M = m, \dim = n$. $r_x := \operatorname{rank}(\mathrm{d}\varphi)_x$.

1. 若对于任意的 $x \in M, r_x = n$. 则称 φ 是一个淹没 (Submersion).
2. 若对于任意的 $x \in M$, $r_x = m$. 则称 φ 是一个浸入 (immersion) .

Proposition 1.4.9

1. 若 φ 是一个淹没, 则 $m \geq n$. 且存在局部坐标系, 使得

$$\tilde{\varphi}\left(x^1, \cdots, x^m\right) = \left(x^1, \cdots, x^n\right)$$

2. 若 φ 是一个浸入, 则 $m \leq n$. 且存在局部坐标, 使得

$$\tilde{\varphi}\left(x^1, \cdots, x^m\right) = \left(x^1, \cdots, x^n, 0, \cdots, 0\right)$$

Definition 1.4.10

设 M, N 是拓扑流形. $\varphi : M \to N$ 是连续映射.

1. 若对于任意的 $x \in M$. 记 $y = \varphi(x)$. 存在 x, y 附近的坐标卡 $\left(U, \varphi_U : U \to \tilde{U}\right), (V, \varphi_v : V \to \tilde{V})$. 使得

$$\tilde{\varphi} = \varphi_V \circ \varphi \circ \varphi_U^{-1} : \tilde{U} \to \tilde{V}$$

   是一个从 $\mathbb{R}^m$ 到 $\mathbb{R}^n$ 的投影. 则称 φ 是一个拓扑淹没.
2. 可以类似地定义出拓扑浸入.

Theorem 1.4.11

设 M 设 C^∞ 流形. N 是拓扑流形. 若 $\varphi : M \to N$ 是拓扑的淹没且满, 则存在唯一的 C^∞ 结构, 使得 φ 是一个 C^∞ 淹没.

Definition 1.4.12

设 $\varphi : M \to N$ 是一个浸入.

1. 若 φ 是单的, 则称 φ 是一个嵌入.

2. 若 φ 是嵌入, 且 $\varphi_M : M \to \varphi(M)$ 是同胚. 则称 φ 是一个正则嵌入.

Theorem 1.4.13 ▶ 正则嵌入的水平集刻画

设 $\varphi : M \to N$ 是正则嵌入. 则对于 $y \in \varphi(M)$. 存在 y 在 N 中的开邻域 V, 以及 V 上的 C^∞ 映射 $F : V \to \mathbb{R}^{n-m}$. 使得 $\varphi(M) \cap V = \{z \in V : F(z) = 0\}$

Theorem 1.4.14 ▶ 正则嵌入的参数刻画

对于 $x \in M, y = \varphi(x) \in \varphi(M)$. 存在 x 在 M 中的开邻域 U, y 在 N 中的开邻域 V. 使得 $\varphi(U) = \varphi(M) \cap V$.

Theorem 1.4.15

设 P 是任一 C^∞ 流形. $\varphi : M \to N$ 是正则嵌入. $u : P \to M$ 是任一映射. 则

$$u\text{是}C^\infty\text{的} \iff \varphi \circ u : P \to N\text{是}C^\infty\text{的}.$$

Definition 1.4.16

若 $\varphi : M \to N$ 是嵌入且满足

$$u\text{是}C^\infty\text{的} \iff \varphi \circ u : P \to N\text{是}C^\infty\text{的}.$$

则称 φ 是拟正则嵌入.

Definition 1.4.17

设 N 是一个 C^∞ 流形. $M \subseteq N$. 若 $i : M \to N$ 是嵌入. 则称 M 是 N 的子流形. 若 i 还是正则或拟正则的. 则称 M 是 N 的正则或拟正则子流形.

Theorem 1.4.18 ▶ 正则值

设 M, N 是 C^∞ 流形. $\varphi : M \to N$ 是 C^∞ 映射. 对于 $y \in N$. 记 $M' = \varphi^{-1}(y)$. 若对任一的 $x \in M'$. $\operatorname{rank}(\mathrm{d}\varphi)_x = n$. 则 M' 是 M 的一个正则子流形, 维数为 $m - n$. 这样的 y 叫 φ 的一个正则值.

Note.

- $m-n$ 衡量的是 M 上有多少维数的信息, 在通过 φ 映射的过程中被压缩掉了.
- y 的正则性, 体现在它的“来源信息”在 M 上是光滑的.
- 正则值 y 给出的正则子流形 M', 就是 M 上那些在映射 φ 下被压缩成 y 的信息构成的子流形.

Example 1.4.19

1. $\mathrm{Mat}(n,\mathbb{R}) \simeq \mathbb{R}^2$ 是 n^2 维流形.
2. $\mathrm{GL}(n,\mathbb{R}) = \{M \in \mathrm{Mat}(n,\mathbb{R}) : \det M \neq 0\}$ 作为 $\mathrm{Mat}(n,\mathbb{R})$ 的一个开子集也是一个 n^2 维流形.
3. $\mathrm{SL}(n,\mathbb{R}) = (M \in \mathrm{Mat}(n,\mathbb{R})) : \det M = 1$

$$\begin{aligned} \det : \mathrm{Mat}(n,\mathbb{R}) &\to \mathbb{R} \\ M &\mapsto \det M \end{aligned}$$

   $SL(n,\mathbb{R}) = \det^{-1}(I)$. 取一个 $M \in \mathrm{Mat}(n,\mathbb{R})$, $\det M = 1$. 要证明 $(\mathrm{d}(\det))_M : \mathbb{R}^{n^2} \to \mathbb{R}$. 按分量求偏导, 得到

$$\frac{\partial}{\partial a_{ij}}(\det M) = A_{ij}(\text{代数余子式})$$

   由于 $\det M = 1$. 至少有一个 $A_{ij} \neq 0$. 从而 $(\mathrm{d}(\det))_M$ 是满的.
4. 定义 $P : \mathrm{Mat}(n,\mathbb{R}) \to \mathrm{SMat}(n,\mathbb{R})$(对称矩阵) $\simeq \mathbb{R}^{\frac{n(n+1)}{2}}$.

$$P(A) := A^{\top}A$$

   则 $O(n,\mathbb{R}) = P^{-1}(I)$. 对于 $A \in O(n,\mathbb{R})$.

$$(\mathrm{d}P)_A : \mathrm{Mat}(n,\mathbb{R}) \to \mathrm{SMat}(n,\mathbb{R})$$

   与上例不同, 这里用分量的形式去计算 Jacobi 是看不清楚的, 不过由于 P 本身足够简单, 我们可以用微分最原始的含义来观察. 考虑

$$P(A+\varepsilon B) - P(A) = \varepsilon(\mathrm{d}P)_A(B) + O(\cdots)$$

   带入得到

$$\left(A^T + \varepsilon B^T\right)(A+\varepsilon B) - A^{\top}A = \varepsilon\left(B^{\top}A + A^{\top}B\right) + O(\varepsilon)$$

于是 $(\mathrm{d}P)_A(B) = B^\top A + A^\top B$. 只需要证明这个关于 B 的映射是满的. 为此, 解方程

$$A^\top B + B^\top A = C$$

取 $B = \frac{1}{2}AC$, 利用 A, C 对称, 带入得到

$$\frac{1}{2}A^\top AC + \frac{1}{2}C^\top A^\top A = \frac{1}{2}\left(C + C^\top\right) = C$$

故 $O(n, \mathbb{R})$ 是 $\mathrm{Mat}(n, \mathbb{R})$ 的一个 $\frac{n(n-1)}{2}$ 维正则子流形. 注意到对于 $A \in O(n, \mathbb{R})$, $\det A = 1$ 或 $\det A = -1$, 可以窥见事实上 $O(n, \mathbb{R})$ 有两个连通分支.

1.5 Frobenius 定理

Definition 1.5.1 ▶ C^∞ 分布

在光滑流形 M 上, 一个 k-维 C^∞ 分布 (smooth distribution) 是一个映射 $\Delta : M \to \bigcup_{p\in M} \mathrm{Gr}(k, T_pM)$, 这里

$$\mathrm{Gr}(k, V) = \{W \subseteq V \mid W \text{ 是} k\text{-维子空间}\}$$

其中:
- 对于每个 $p \in M$, Δ_p 是切空间 T_pM 的一个 k-维子空间.
- Δ 是光滑的, 即它对应于切丛 TM 的一个光滑子向量丛: 局部存在光滑向量场 $X_1, \dots, X_k$ 生成 $\Delta_p = \mathrm{span}\{X_1(p), \dots, X_k(p)\}$.

Definition 1.5.2

设 $\mathscr{L}$ 是 N 上的一个分布.
- 设 $\mathscr{L}$ 是一个 m 维分布. M 是 N 的一个 m 维子流形. 若对于 $x \in M$. $T_xM = L_x$, 则称 M 是 $\mathscr{L}$ 的一个积分子流形.
- 若对于任意的 $x \in N$. 都存在一个过 x 的 $\mathscr{L}$ 的积分子流形. 则称 $\mathscr{L}$ 是可积的.
- 若 $[\mathscr{L}, \mathscr{L}] \subseteq \mathscr{L}$, 则称 $\mathscr{L}$ 是对合的 (involutive) .

Remark.
由于 Lie bracket 是封闭的, 从而可积一定是对合的.

Theorem 1.5.3 ▸ 局部 Frobenius

设 $\mathscr{L}$ 是 N 上的一个分布. 若 $\mathscr{L}$ 是对合的, 则 $\mathscr{L}$ 是可积的.

Theorem 1.5.4 ▸ 整体 Frobenius

设 N 是一个 n 维 C^∞ 流形. $\mathscr{L}$ 是 N 上的一个 m 维对合分布. 则过 N 上任一点 p, 存在唯一的 $\mathscr{L}$ 的极大积分子流形 M_p. $\mathscr{L}$ 的任何积分子流形都是 N 的拟正则子流形, 并且是某一个极大积分子流形的开子流形.

Proof sketch.
对 $p \in N$,

$$M_p = \{q \in N : \text{存在一条分片光滑的}\mathscr{L}\text{的积分曲线, 连接}p\text{和}q\}$$

参考 Warner 的 GTM94

2 Basic Concepts of Lie Algebra

2.1 Definitions and first examples

Definition 2.1.1

A vector space L over a field $\mathbb{F}$, with an operation $L \times L \to L$, denoted $(x, y) \mapsto [xy]$ and called the ***bracket*** or ***commutator*** of x and y, is called a ***Lie algebra*** over $\mathbb{F}$ if the following axioms are satisfied:
(L1) The bracket operation is bilinear.
(L2) $[xx] = 0$ for all $x \in L$.
(L3) $[x[yz]] + [y[zx]] + [z[xy]] = 0 \quad (x, y, z \in L)$.

Remark.
- (L1) and (L2), applied to $[x + y, x + y]$ imply anticommutativity : (L2′) $[xy] = [-yx]$.
- Conversely, if $\operatorname{char} \mathbb{F} \neq 2$, it is clear that (L2′) will imply (L2).

Definition 2.1.2

We define a ***Lie subalgebra*** of L to be a vector subspace $K \subseteq L$ such that

$$[x, y] \in K \quad \text{for all } x, y \in K.$$

Example 2.1.3

Suppose that V is a finite-dimensional vector space over F. Write $\mathfrak{gl}(V)$ for the set of all linear maps from V to V. This is again a vector space over F, and it becomes a Lie algebra, known as the ***general linear algebra***, if we define the Lie bracket $[-, -]$ by

$$[x, y] := x \circ y - y \circ x \quad \text{for } x, y \in \mathfrak{gl}(V),$$

where $\circ$ denotes the composition of maps.

2.2 Lie algebras of derivations

Definition 2.2.1

- By an ***F-algebra*** (not necessarily associative) we simply mean a vector space $\mathfrak{A}$ over F endowed with a bilinear operation $\mathfrak{A} \times \mathfrak{A} \to \mathfrak{A}$, usually denoted by juxtaposition (unless $\mathfrak{A}$ is a Lie algebra, in which case we always use the bracket).
- By a ***derivation*** of $\mathfrak{A}$ we mean a linear map $\delta : \mathfrak{A} \to \mathfrak{A}$ satisfying the familiar product rule $\delta(ab) = a\delta(b) + \delta(a)b$ for each $n \in \mathbb{Z}$.

2.3 Ideals and homomorphism

2.3.1 Ideals

Definition 2.3.1

A subspace I of a Lie algebra L is called an ***ideal*** of L if $x \in L, y \in I$ together imply $[xy] \in I$.

Remark.
由于 $[x, y] = -[y, x]$, 理想不分左右.

2.3.2 Homomorphisms and representations

Definition 2.3.2

- A linear transformation $\phi : L \to L'$ (L, L' Lie algebras over F) is called a ***homomorphism*** if $\phi([xy]) = [\phi(x)\phi(y)]$, for all $x, y \in L$.
- ϕ is called a ***monomorphism*** if $\mathrm{Ker}\phi = 0$, an ***epimorphism*** if $\mathrm{Im}\phi = L'$, an ***isomorphism*** if it is both mono- and epi-.

Definition 2.3.3

If L_1 and L_2 are Lie algebras over a field F, then we say that a map $\varphi : L_1 \to L_2$ is a ***homomorphism*** if φ is a linear map and

$$\varphi([x, y]) = [\varphi(x), \varphi(y)] \quad \text{for all } x, y \in L_1.$$

Notice that in this equation the first Lie bracket is taken in L_1 and the second Lie bracket is taken in L_2. We say that φ is an ***isomorphism*** if φ is also bijective.

Example 2.3.4

An extremely important homomorphism is the ***adjoint homomorphism***. If L is a Lie algebra, we define

$$\operatorname{ad} : L \to \mathfrak{gl}(L)$$

by $(\operatorname{ad} x)(y) := [x, y]$ for $x, y \in L$. It follows from the bilinearity of the Lie bracket that the map $\operatorname{ad} x$ is linear for each $x \in L$. For the same reason, the map $x \mapsto \operatorname{ad} x$ is itself linear. So to show that ad is a homomorphism, all we need to check is that

$$\operatorname{ad}([x, y]) = \operatorname{ad} x \circ \operatorname{ad} y - \operatorname{ad} y \circ \operatorname{ad} x \quad \text{for all } x, y \in L;$$

this turns out to be equivalent to the Jacobi identity. The kernel of ad is the centre of L.

Proof. We have from Jacobi identity that

$$\begin{aligned} \operatorname{ad}([x, y])(z) &= [[x, y], z] \\ &= [x, [y, z]] + [y, [z, x]] \\ &= [x, \operatorname{ad} y(z)] - [y, \operatorname{ad} x(z)] \\ &= \operatorname{ad} x \circ \operatorname{ad} y(z) - \operatorname{ad} y \circ \operatorname{ad} x(z) \end{aligned}$$

□

3 Solvable Lie Algebras

3.1 Solvable Lie Algebras

Lemma 3.1.1

Suppose that I is an ideal of L. Then L/I is abelian if and only if I contains the derived algebra L'.

Proof. L/I is abelian, if and only if for each $x, y \in L$,

$$[x+I, y+I] = [x,y] + I = 0 + I \iff [x,y] \in I$$

That is L/I is abelian, if and only if $L' = [L,L] \subseteq I$. □

Definition 3.1.2

This lemma tells us that the derived algebra L' is the smallest ideal of L with an abelian quotient. By the same argument, the derived algebra L' itself has a smallest ideal whose quotient is abelian, namely the derived algebra of L', which we denote $L^{(2)}$, and so on. We define the ***derived series*** of L to be the series with terms

$$L^{(1)} = L' \quad \text{and} \quad L^{(k)} = [L^{(k-1)}, L^{(k-1)}] \text{ for } k \geq 2.$$

Then $L \supseteq L^{(1)} \supseteq L^{(2)} \supseteq \ldots$. As the product of ideals is an ideal, $L^{(k)}$ is an ideal of L (and not just an ideal of $L^{(k-1)}$).

Definition 3.1.3

The Lie algebra L is said to be ***solvable*** if for some $m \geq 1$ we have $L^{(m)} = 0$.

Lemma 3.1.4

If L is a Lie algebra with ideals

$$L = I_0 \supseteq I_1 \supseteq \ldots \supseteq I_{m-1} \supseteq I_m = 0$$

such that I_{k-1}/I_k is abelian for $1 \leq k \leq m$, then L is solvable.

Note.

- 我们把这样的理想链视作过滤掉非交换性的多层滤网，每次都把可交换的元（相当于粗颗粒）保留，将不能交换的程度（细颗粒）过滤下去.
- 导出列实现对不可交换性的精准过滤和采集.
- 普通的理想链是可能更粗的滤网，它能承载所有上一层级的非交换性，但是可能混入可交换性的细颗粒.
- 这个引理相当于是说，如果我们通过可能更不精细的手法将所有的非交换性过滤干净，那么原有的精确过滤也一定能过滤干净.

Proof. We shall inductively shows that $L^{(k)}$ is contained in I_k, then by taking $k = m$, we get $L^{(m)} = 0$.

Since $L/I_1 = I_0/I_1$ is abelian, $I_1 \supseteq L'$. From inductive hypothesis, we know that $L^{(k-1)} \subseteq I_{k-1}$, since I_{k-1}/I_k is abelian, that is

$$[I_{k-1}/I_k, I_{k-1}/I_k] = 0$$

which is equivalent to

$$[I_{k-1}, I_{k-1}] \subseteq I_k$$

Since $L^{(k-1)} \subseteq I_{k-1}$, we have

$$L^{(k)} = [L^{(k-1)}]$$

□

Proposition 3.1.5

Suppose that $\varphi : L_1 \to L_2$ is a surjective homomorphism of Lie algebras. Show that

$$\varphi(L_1^{(k)}) = (L_2)^{(k)}.$$

Proof. For $k = 0$,

$$\varphi(L_1) = L_2$$

Inductively,

$$\varphi\left(L_1^{(k)}\right) = \varphi\left(\left[L_1^{(k-1)}, L_1^{(k-1)}\right]\right) = \left[\varphi\left(L_1^{(k-1)}\right), \varphi\left(L_1^{(k-1)}\right)\right] = \left[L_2^{(k-1)}, L_2^{(k-1)}\right] = L_2^{(k)}$$

□

Lemma 3.1.6

Let L be a Lie algebra.
(a) If L is solvable, then every subalgebra and every homomorphic image of L are solvable.
(b) Suppose that L has an ideal I such that I and L/I are solvable. Then L is solvable.
(c) If I and J are solvable ideals of L, then $I + J$ is a solvable ideal of L.

(a) Omitted.

(b)

> *Note.*
> 商去 I 是可解的，当且仅当各层级的非交换性可以被 I 给兜住.

From the above proposition, we have $(L/I)^{(k)} = \left(L^{(k)} + I\right)/I$. If L/I is solvable, then $L^{(m)} \subseteq I$ for some m. We have for some k,

$$\left(L^{(m)}\right)^{(k)} \subseteq I^{(k)} = 0$$

Alternatively, it follows from (a) that $L^{(m)}$ is sovable.

$$L^{(m+k)} = \left(L^{(m)}\right)^{(k)} = 0$$

(c) $(I + J)/I \simeq J/I \cap J$ is sovable by (a). And since I is sovable , $I + J$ is sovable by (b).

4 Subalgebra of the General Linear Algebra

4.1 Nilpotent Maps

Lemma 4.1.1

Let $x \in L$. If the linear map $x : V \to V$ is nilpotent, then $\operatorname{ad} x : L \to L$ is also nilpotent.

Proof. Take any $y \in L$, and expand

$$(\operatorname{ad} x)^m (y) = [x, [x, \cdots [x, y] \cdots]]$$

which is of the form

$$x^i y x^j, \quad i + j = m$$

Suppose that $x^r = 0$, taking $m = 2r$, we have $(\operatorname{ad} x)^m (y) = 0$ for all $y \in L$. □

Lemma 4.1.2

Suppose that A is an ideal of a Lie subalgebra L of $\mathfrak{gl}(V)$. Let

$$W = \{v \in V : a(v) = 0 \text{ for all } a \in A\}.$$

Then W is an L-invariant subspace of V.

Remark.
我们熟知, 如果每个 a 和 y 可交换, 则 W 一定是 L 不变的. 因此 W 距离成为 L 不变子空间, 只在于是否在 $[a, y]$ 下保持, 这也是为什么我们要假设 A 是一个理想.

Proof. Take $w \in W$ and $y \in L$, we must show that $a\,(yw) = 0$ for all a . But $ay = ya + [a, y]$, we have

$$a\,(yw) = y\,(aw) + [a, y]\,(w)$$

And $y\,(aw) = 0$ for $aw = 0$; $[a, y]\,(w) = 0$ for $[a, y] \in A$. □

4.2 Weights

Definition 4.2.1

A ***weight*** for a Lie subalgebra A of $\mathfrak{gl}(V)$ is a linear map $\lambda : A \to F$ such that
$$V_\lambda := \{v \in V : a(v) = \lambda(a)v \text{ for all } a \in A\}$$
is a non-zero subspace of V.

4.3 Invariance Lemma

Lemma 4.3.1

Suppose that A is an ideal of a Lie subalgebra L of $\mathfrak{gl}(V)$. Let
$$W = \{v \in V : a(v) = 0 \text{ for all } a \in A\}.$$
Then W is an L-invariant subspace of V.

Lemma 4.3.2 ▶ Invariance Lemma

Assume that F has characteristic zero. Let L be a Lie subalgebra of $\mathfrak{gl}(V)$ and let A be an ideal of L. Let $\lambda : A \to F$ be a weight of A. The associated weight space
$$V_\lambda = \{v \in V : av = \lambda(a)v \text{ for all } a \in A\}$$
is an L-invariant subspace of V.

5 Engel's Theorem and Lie's Theorem

5.1 Engel's Theorem

Lemma 5.1.1

Suppose that L is a Lie subalgebra of $\mathfrak{gl}(V)$, where V is non-zero, such that every element of L is a nilpotent linear transformation. Then there is some non-zero $v \in V$ such that $xv = 0$ for all $x \in L$.

Proof sketch.
对维数归纳
对于任意 L 的真子代数 A, 模 A 的伴随:

$$\begin{aligned} \varphi : A &\to \mathfrak{gl}(L/A) \\ a &\mapsto (x + A \mapsto [a, x] + A) \end{aligned}$$

我们看到 $\varphi(A)$ 的公共 0-特征向量, 就是关于 A 代数封闭的向量.
根据归纳假设, $\varphi(A)$ 有非零公共 0-特征向量 y, 从而 $y \notin A$ 且 y 是关于 A 代数封闭的. 这就意味着 A 可以成为一个更大的代数 $\tilde{A} = A \oplus \operatorname{span}\{y\}$ 的理想. 如果我们取 A 是极大的, 那么 $\tilde{A} = L$. 从而 A 成为 L 的一个余维数为 1 的理想.
一个引理是, L 的理想 A 的零子空间 W 是 L 不变的子空间. 这就意味着当我们寻找 $L = A \oplus \operatorname{span}\{y\}$ 的公共零特征向量时, 可以从 A 的零子空间 W 中, 寻找 $\operatorname{span}\{y\}$ 的公共零特征向量, 这是可行的, 因为 $\operatorname{span}\{y\}$ 也是一组幂零的线性变换.

Proof. We induct on the dimension of L. Let A be a maximal subalgebra of L. Define a map

$$\begin{aligned} \varphi : A &\to \mathfrak{gl}(L/A) \\ a &\mapsto (x + A \mapsto [a, x] + A) \end{aligned}$$

It is straightforward to verify that φ is a Lie algebra homomorphism. Thus $\varphi(A)$ is a Lie algebra . For each $a \in A$, since a is nilpotent, ad a is as well, hence $\varphi(a)$ is nilpotent. We know from inductive hypothesis that there exists nonzero $y + A \in L/A$ such that $0 = \varphi(a)(y + A), \forall a \in A$. That is $y \notin A$ and

$[A, y] \subseteq A$. By defining $\tilde{A} := A \oplus \operatorname{span}\{y\}$, we know A is an ideal of $\tilde{A}$. Since A is maximal, there must be $\tilde{A} = L$, A is an ideal of L with codimension 1.

A lemma gives that the space W

$$W = \{v \in V : a(v) = 0, \forall a \in A\}$$

is L-invariant. Specifically, $y(W) \subseteq W$. $y|_W$ is nilpotent, there exist nonzero $v \in W$ such that $y(v) = y|_W(v) = 0$. Then for each element of L, we can write it as $a + ky$ for some $k \in F$. Then

$$(a + ky)(v) = a(v) + ky(v) = 0$$

which completes the proof. □

Theorem 5.1.2 ▸ Engel's Theorem

Let V be a vector space. Suppose that L is a Lie subalgebra of $\mathfrak{gl}(V)$ such that every element of L is a nilpotent linear transformation of V. Then there is a basis of V in which every element of L is represented by a strictly upper triangular matrix.

Proof. We induct on the dimension of V. By Lemma 5.1.1, there exists $w \in V$ such that $x(w) = 0, \forall x \in L$. Let $W = \operatorname{span}\{w\}$. Then every x induces a linear transformation $\bar{x} : V/W \to V/W$, $\bar{x}(v + W) = x(v) + W$. It is easily to check that the map

$$L \to \mathfrak{gl}(V/W), \quad x \mapsto \bar{x}$$

is a Lie homomorphism.

The image of L under this homomorphism is a subalgebra of $\mathfrak{gl}(V/W)$, which satisfies the hypothesis of Engel's Theorem. Moreover $\dim(V/W) = \dim V - 1$. We know from inductive hypothesis that there is a basis of V/W, say $\{v_i + W\}_{2 \leq i \leq n}$ in which every element of L is represented by a strictly upper triangular matrix. That is equivalent to for each $x \in L$,

$$\bar{x}\left(v_j + W\right) \in \operatorname{span}\left\{v_2 + W, \cdots, v_{j-1} + W\right\}, \quad \forall 3 \leq j \leq n,$$

and

$$\bar{x}(v_2 + W) = 0 + W$$

By taking $v_1 = w_{,}$, it follows that

$$x(v_1) = 0, \quad x(v_2) \in \operatorname{span}\{v_1\}$$

$$x(v_j) \in \operatorname{span}\{v_1, v_2, \cdots, v_{j-1}\}, \quad \forall 3 \leq j \leq n$$

That is every element of L is represented by a strictly upper triangular matrix under the basis $\{v_1, v_2, \cdots, v_n\}$. □

Definition 5.1.3

Recall that a Lie algebra is said to be nilpotent if for some $m \geq 1$ we have $L^m = 0$ or, equivalently, if for all $x_0, x_1, \ldots, x_m \in V$ we have

$$[x_0, [x_1, \ldots, [x_{m-1}, x_m] \ldots]] = 0.$$

Theorem 5.1.4 ▸ Engel's Theorem, second version

A Lie algebra L is nilpotent if and only if for all $x \in L$ the linear map $\operatorname{ad} x : L \to L$ is nilpotent.

Proof. The 'only if' is straightforward. For the 'if' part , if all ad x is nilpotent, the algebra

$$\operatorname{ad}(L) = \{\operatorname{ad} x : x \in L\}$$

is a subalgebra of $\mathfrak{gl}(L)$ satisfying the hypothesis of Engel's Theorem. it follows from Engel's Theorem that there exists a basis of L say $\{e_1, \cdots, e_n\}$, such that

$$[L, e_i] \subseteq \operatorname{span}\{e_1, \cdots, e_{i-1}\}$$

Denote L_j by $\operatorname{span}\{e_1, \cdots, e_j\}$, then

$$[L, L_j] \subseteq L_{j-1}, j \geq 2, \quad [L, L_1] = 0$$

We have

$$L^0 = L = L_n$$

$$L^1 = [L, L] \subseteq L_{n-1}$$

Inductively, we have

$$L^k = [L, L^{k-1}] \subseteq [L, L_{n-k+1}] \subseteq L_{n-k}$$

Finally ,we have

$$L^n = 0$$

L is nilpotent. □

5.2 Lie's Theorem

Proposition 5.2.1

Let V be a non-zero complex vector space. Suppose that L is a solvable Lie subalgebra of $\mathfrak{gl}(V)$. Then there is some non-zero $v \in V$ which is a simultaneous eigenvector for all $x \in L$

Proof sketch.

- 一个李代数 L 的子代数 A 如果包含了 L', 则 A 是 L 的一个理想.
- 如果 L 是可解的, 则 L' 是真子空间, 这意味着我们可以构造 L 的余一维的理想.
- 归纳假设给出 A 的一个特征向量, 从而对应的 weight 子空间非空.
- 理想的 Weight 子空间是 L-不变的, 特别地是 v-不变的.
- 由于空间是 $\mathbb{C}$ 上的, 在 Weight 子空间上存在特征向量 v, 它就是公共的特征向量.

Theorem 5.2.2 ▸ Lie's Theorem

Let V be an n-dimensional complex vector space and let L be a solvable Lie subalgebra of $\mathfrak{gl}(V)$. Then there is a basis of V in which every element of L is represented by an upper triangular matrix.

Proof sketch.
证明完全类似于 Engel 定理的证明.

6 Representation of $\mathfrak{sl}(2, \mathbb{C})$

6.1 The Modules V_d

6.2 Classifying the Irreducible $\mathfrak{sl}(2, \mathbb{C})$-Modules

Lemma 6.2.1

Suppose that V is an $\mathfrak{sl}(2, \mathbb{C})$-module and $v \in V$ is an eigenvector of h with eigenvalue λ.

- Either $e \cdot v = 0$ or $e \cdot v$ is an eigenvector of h with eigenvalue $\lambda + 2$.
- Either $f \cdot v = 0$ or $f \cdot v$ is an eigenvector of h with eigenvalue $\lambda - 2$.

Lemma 6.2.2

Let V be a finite-dimensional $\mathfrak{sl}(2, \mathbb{C})$-module. Then V contains eigenvector w for h such that $e \cdot w = 0$.

Theorem 6.2.3

If V is finite-dimensional irreducible $\mathfrak{sl}(2, \mathbb{C})$-module, then V is isomorphic to one of the V_d.

Corollary 6.2.4

If V is a finite-dimensional representation of $\mathfrak{sl}(2, \mathbb{C})$ and $w \in V$ is an h-eigenvector such that $e \cdot w = 0$, then $h \cdot w = dw$ for some non-negative interger d and the submodule of V generated by w is isomorphic to V_d.

6.3 Weyl's Theorem

Theorem 6.3.1

Let L be a complex semisimple Lie algebra. Every finite-dimensional representation of L is completely reducible.

7 Cartan Criteria

7.1 Jordan Decomposition

Lemma 7.1.1

Let x be a linear transformation of the complex vector space V. Suppose that x has Jordan decomposition $x = d + n$, where d is diagonalisable, n is nilpotent, and d and n commute.

(a) There is a polynomial $p(X) \in \mathbb{C}[X]$ such that $p(x) = d$.

(b) Fix a basis of V in which d is diagonal. Let $\bar{d}$ be the linear map whose matrix with respect to this basis is the complex conjugate of the matrix of d. There is a polynomial $q(X) \in \mathbb{C}[X]$ such that $q(x) = \bar{d}$.

Exercise 7.1.2

Let V be a vector space, and suppose that $x \in \mathfrak{gl}(V)$ has Jordan decomposition $d + n$. Show that $\operatorname{ad} x : \mathfrak{gl}(V) \to \mathfrak{gl}(V)$ has Jordan decomposition $\operatorname{ad} d + \operatorname{ad} n$.

Proof. Only need to show that $\operatorname{ad} d$ is semisimple and $\operatorname{ad} n$ is nilpotent, then the proposition follows from the linearity of ad and the uniqness of Jordan decomposition.

- Suppose that $e_1, \cdots, e_n$ is the basis in which d is represented diagonal, say $\operatorname{diag}(\lambda_1, \cdots, \lambda_n)$. Let $E_{ij} \in \mathfrak{gl}(V)$ be the maps that mapping e_j to e_i, that is $E_{ij}(e_k) = \delta_j^k e_i$, $k = 1, \cdots, n$. Then

$$\begin{aligned}
\operatorname{ad}(d)\left(E_{ij}\right)(e_k) &= \left[d, E_{ij}\right](e_k) \\
&= \left(dE_{ij} - E_{ij}d\right)(e_k) \\
&= d\left(E_{ij}e_k\right) - E_{ij}(de_k) \\
&= d\left(\delta_j^k e_i\right) - E_{ij}(\lambda_k e_k) \\
&= \delta_j^k \lambda_i e_i - \lambda_k \delta_j^k e_i \\
&= (\lambda_i - \lambda_k)\delta_j^k e_i = \left(\lambda_i - \lambda_j\right) E_{ij}
\end{aligned}$$

Which implies that E_{ij} is a eigenvector for $\operatorname{ad}(d)$ for each $i, j = 1, \cdots, n$. $\operatorname{ad} n$ is diagonal under the basis $E_{ij}, i, j = 1, \cdots, n$

- Since n is nilpotent, it follows that $\operatorname{ad} n$ is nilpotent.
- It follows from Jacobi identity that

$$(\operatorname{ad} d) \circ (\operatorname{ad} n)(y) = [d, [n, y]] = -[y, [d, n]] - [n, [y, d]]$$

Since d, n commutes, $[d, n] = 0$. Thus $\operatorname{ad} n$ and $\operatorname{ad} d$ commute.

□

7.2 Testing for Solvability

Exercise 7.2.1

Let V be a complex vector space, L be a Lie subalgebra of $\mathfrak{gl}(V)$. Suppose that L is solvable. Use Lie's Theorem to show that there is a basis of V in which every element of L' is represented by a strictly upper triangular matrix. Conclude that $\operatorname{tr} xy = 0$ for all $x \in L$ and $y \in L'$.

Remark.
$\mathbb{C}$ can replaced by $\mathbb{F}$ such that $\operatorname{char} \mathbb{F} = 0, \bar{\mathbb{F}} = \mathbb{F}$.

Proof. From Lie's Theorem ,there is a basis of V in which every element of L is upper triangular , say $\{e_1, \cdots, e_n\}$. Take $z \in L', z = [x, y], x, y \in L$. Then

$$z(e_1) = [x, y](e_1) = x(y(e_1)) - y(x(e_1)) = 0$$

Since e_1 is simultaneous an eigenvector of x and y. Let $L_k = \operatorname{span}\{e_1, \cdots, e_k\}$, then

$$z(e_k) = x(y(e_k)) - y(x(e_k))$$

Suppose ,

$$y(e_k) = \sum_{j=1}^{k} y_i e_j, \quad x(e_k) = \sum_{j=1}^{k} x_i e_j$$

then

$$\begin{aligned} x(y(e_k)) + \operatorname{span}\{e_1, \cdots, e_{k-1}\} &= x(y_i e_k) + \operatorname{span}\{e_1, \cdots, e_{k-1}\} \\ &= x_i y_i e_k + \operatorname{span}\{e_1, \cdots, e_{k-1}\} \end{aligned}$$

Similarly,

$$y\left(x\left(e_k\right)\right)+\operatorname{span}\left\{e_1,\cdots,e_{k-1}\right\}=x_i y_i e_k+\operatorname{span}\left\{e_1,\cdots,e_{k-1}\right\}$$

Thus

$$z\left(e_k\right)\in\operatorname{span}\left\{e_1,\cdots,e_{k-1}\right\}$$

It follows that z is nilpotent. Thus every element of L' is represented by a strictly upper triangular matrix.

The matrix of x and y under the basis $\{e_1,\cdots,e_n\}$ are upper triangular matrix and strictly upper triangular matrix , repectively , say A, B. Then

$$\operatorname{tr} xy=\operatorname{tr}(AB)=\sum_{k=1}^{n} a_{ik}b_{kj}=0$$

Since for each $i>k$, $a_{ik}=0$, and for each $i\leq k$, $b_{kj}\leq 0$. □

Proposition 7.2.2

Let V be a complex vector space and let L be a Lie subalgebra of $\mathfrak{gl}(V)$. If $\operatorname{tr} xy=0$ for all $x,y\in L$, then L is solvable.

Proof. We shall show that every element of L' is nilpotent, then it follows from Engel's Theorem that L' is nilpotent, thus L is solvable.

Take any element $x\in L'$. Suppose that x has a Jordan decomposition $x=d+n$. We choose a basis in which d is diagonal $d=\operatorname{diag}(\lambda_1,\cdots,\lambda_n)$ and n is strictly upper triangular. We only need to show that

$$\sum_{i=1}^{n}\lambda_i\bar{\lambda}_i=0$$

$$\begin{aligned}\operatorname{tr}\left(\bar{d}x\right)&=\operatorname{tr}\left(\bar{d}d+\bar{d}n\right)=\operatorname{tr}\left(\bar{d}d+\bar{d}n\right)\\&=\operatorname{tr}\left(\bar{d}d\right)+0\\&=\sum_{i=1}^{n}\lambda_i\bar{\lambda}_i\end{aligned}$$

Since $x \in L'$,we write $x = [y, z]$, $y, z \in L'$.

$$\begin{aligned} \operatorname{tr}\left(\bar{d}x\right) &= \operatorname{tr}\left(\bar{d}\left[y,z\right]\right) \\ &= \operatorname{tr}\left(\bar{d}yz - \bar{d}zy\right) = \operatorname{tr}\left(\bar{d}yz - y\bar{d}z\right) \\ &= \operatorname{tr}\left(\left[\bar{d}, y\right] z\right) \end{aligned}$$

Note.
这里我们折腾 $\operatorname{ad}\bar{d}$ 而非 $\bar{d}$ 的意义在于, 多项式对不变性的保持需要我们对结合代数封闭, 而李代数不一定保持结合代数的封闭性.

Thus to show $d = 0$, we only need to show that $\left[\bar{d}, y\right] \in L$, that is $\operatorname{ad}_{\bar{d}}(y) \in L$. $\operatorname{ad} x$ has Jordan decomposition, $\operatorname{ad} x = \operatorname{ad} d + \operatorname{ad} n$. Then from Lemma 7.1.1 , there exists $q(X) \in \mathbb{C}[X]$ such that

$$q(\operatorname{ad} x) = \operatorname{ad} \bar{d}$$

Since L is $\operatorname{ad} x$- invariant, so dose $q(\operatorname{ad} x)$ and $\operatorname{ad}\bar{d}$, which completes the proof. □

Theorem 7.2.3

Let L be a complex Lie algebra. Then L is solvable if and only if $\operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y) = 0$ for all $x \in L$ and all $y \in L'$.

Note.
由于 L 的 sovability 等价于任意阶导出子代数 $L^{(k)}$ 的 sovability, 因此命题中的

$$x \in L, \quad y \in L'$$

对于任意固定的 $k \geq 1, l \geq 0$ 可以替换为

$$x \in L^{(k)}, \quad y \in L^{(l)},$$

Proof. Suppose L is sovable. Then $\operatorname{ad}(L) \subseteq \mathfrak{gl}(L)$ is sovable. Then it follows from exercise 7.2.1 that $\operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y) = 0$ for all $x \in L, y \in L'$.

Conversly, it follows from proposition 7.2.2 that $\operatorname{ad}(L')$ is sovable, then so dose L' and L. □

7.3 Killing Form

Definition 7.3.1

Let L be a complex Lie algebra. The Killing form on L is the symmetric bilinear form defined by

$$\kappa(x, y) := \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y), \quad \text{for } x, y \in L$$

Proposition 7.3.2

$$\kappa([x, y], z) = \kappa(x, [y, z]).$$

Theorem 7.3.3 ▸ Cartan's First Criterion

The complex Lie algebra L is solvable if and only if $\kappa(x, y) = 0$ for all $x \in L$ and $y \in L'$.

Proof. Just a restatement of theorem 7.2.3 . □

Lemma 7.3.4

Suppose that L is a Lie algebra and I is an ideal of L. We write κ for the Killing form on L and κ_I for the Killing form on I, considered as a Lie algebra in its own right. If $x, y \in I$, then $\kappa_I(x, y) = \kappa(x, y)$.

Proof. We expand a basis of I to a basis of L. Since

$$\operatorname{ad} x\,(L) \subseteq [x, L] \subseteq I$$

Then $\operatorname{ad} x$ is represented of the form

$$\begin{pmatrix} A_x & B_x \\ O & O \end{pmatrix}$$

$\operatorname{ad} y$ is represented of the form

$$\begin{pmatrix} A_y & B_y \\ O & O \end{pmatrix}$$

We have

$$\begin{aligned}\kappa(x,y) &= \operatorname{tr}\left(\begin{pmatrix} A_x & B_x \\ O & O \end{pmatrix}\begin{pmatrix} A_y & B_y \\ O & O \end{pmatrix}\right) \\ &= \operatorname{tr}\left(\begin{pmatrix} A_xA_y & A_xB_Y \\ O & O \end{pmatrix}\right) \\ &= \operatorname{tr}(A_xA_y) \\ &= \kappa_I(x,y)\end{aligned}$$

□

7.3.1 Testing for Semisimplicity

Definition 7.3.5 ▸ perpendicular space

- Let β be a symmetric bilinear form on a finite-dimensional complex vector space V. If S is a subset of V, we define the ***perpendicular space*** to S by

$$S^\perp = \{x \in V : \beta(x,s) = 0 \text{ for all } s \in S\}.$$

- We say that β is ***non-degenerate*** if $V^\perp = 0$; that is, there is no non-zero vector $v \in V$ such that $\beta(v,x) = 0$ for all $x \in V$.

Proposition 7.3.6

If β is non-degenerate and W is a vector subspace of V, then

$$\dim W + \dim W^\perp = \dim V.$$

Proof. Define a linear map $\varphi : V \to V^*$,

$$\varphi(v) \mapsto (u \mapsto \beta(u,v))$$

Then

$$\ker\varphi = \{v \in V : \beta(u,v) = 0, \forall u \in V\} = V^\perp = 0$$

Since V is finite-dimensional , we have $\dim V = \dim V^*$. Then φ is a isomorphism. Define $W^0 \subseteq V^*$:

$$W^0 = \{\lambda \in V^* : \lambda(w) = 0, \forall w \in W\}$$

We have
$$\varphi\left(W^{\perp}\right) = W^0, \quad \dim W^{\perp} = \dim W^0$$
Let $i : W \hookrightarrow V$ the inclusion map , for each $\lambda^* \in V^*$, we define a map $i^* : V^* \to W^*$
$$i^*(\lambda)(w) = \lambda(i(w))$$
Then $\operatorname{Im} i^* = W^*$, $\ker i^* = W^0$, we have
$$\dim W^0 + \dim W^* = \dim V^*$$
Then it follows that
$$\dim W^{\top} + \dim W^* = \dim V^* \implies \dim W^{\perp} + \dim W = \dim V$$
Since W, V have finite dimension. $\square$

Exercise 7.3.7

Suppose I is an ideal of L. Shows that $I^{\perp}$ is an ideal of L.

Proof.
$$I^{\perp} = \{x \in L : \kappa(i, x) = 0, \forall i \in I\}$$
Take $x \in I^{\perp}$, $y \in L$, note that
$$[y, x] \in I^{\perp} \iff \kappa([x, y], i), \quad \forall i \in I$$
Since we have
$$\kappa(x, [y, i]) = 0, \forall y \in L, i \in I$$
Then
$$\kappa([x, y], i) = \operatorname{tr}\left(\left[\operatorname{ad}_x, \operatorname{ad}_y\right] \circ \operatorname{ad}_i\right) = \operatorname{tr}\left(\operatorname{ad}_x \circ \left[\operatorname{ad}_y \circ \operatorname{ad}_i\right]\right) = \kappa(x, [y, i]) = 0$$
For all $i \in I$. That is $[x, y] \in I^{\perp}$. $\left[L, I^{\perp}\right] \subseteq I^{\perp}$. $\square$

Corollary 7.3.8

If L is semisimple, then κ is non-degenerate.

Proof. Other wise , $L^{\perp}$ is an nonzero ideal of L. Since $\left(L^{\perp}\right)' \subseteq L$, we have
$$\kappa\left(L^{\perp}, \left(L^{\perp}\right)'\right) \subseteq \kappa\left(L^{\perp}, L\right) = 0$$

Then it follows from the Cartan First Criteria that $L^{\top}$ is solvable. Then $L^{\perp}$ is a nonzero solvable ideal of L, which is a contradiction to the Semisimplicity of L. $\square$

Theorem 7.3.9 ▸ Cartan's Second Criterion

The complex Lie algebra L is semisimple if and only if the Killing form κ of L is non-degenerate.

7.4 Derivations of Semisimple Lie Algebras

Proposition 7.4.1

If L is a finite-dimensional complex semisimple Lie algebra, then $\operatorname{ad} L = \operatorname{Der} L$.

Proof. For each $x \in L$,

$$\begin{aligned}\operatorname{ad} x\,[y,z] &= [x,[y,z]]\\ &= -[z,[x,y]] - [y,[z,x]]\\ &= [\operatorname{ad} x\,(y)\,,z] + [y,\operatorname{ad} x\,(z)]\end{aligned}$$

$\operatorname{ad} x$ is a derivation. ad is a Lie algebra homomorphism from L to $\operatorname{Der} L$. For each $\delta \in \operatorname{Der} L$,

$$\begin{aligned}[\delta,\operatorname{ad} x]y &= \delta\,[x,y] - \operatorname{ad} x\,(\delta y)\\ &= [\delta x,y] + [x,\delta y] - [x,\delta y]\\ &= \operatorname{ad}\,(\delta x)\,(y)\end{aligned}$$

Thus $[\delta,\operatorname{ad} x] = \operatorname{ad}\,(\delta x)$, the image of ad is an ideal of $\operatorname{Der} L$.

Now we bring in our assumption that L is complex and semisimple. First , note that $\operatorname{ad} : L \to \operatorname{Der} L$ is one-to-one , as $\ker \operatorname{ad} = Z(L) = 0$, so the lie algebra $M \coloneqq \operatorname{ad} L$ is isomorphic to L, therefore it is semisimple as well. We have

$$\dim M + \dim M^{\perp} = \dim \operatorname{Der} L$$

so our aim is to show that $\dim M^{\perp} = 0$. As M is an ideal of $\operatorname{Der} L$, the Killing form κ_M ofr M is the restriction of the Killing form on $\operatorname{Der} L$. By Cartan's Second Criterion, κ_M is non-degenerate, so $M^{\perp} \cap M = 0$, and hence $\left[M^{\top}, M\right] = 0$. Thus if $\delta \in M^{\perp}$, and $\operatorname{ad} x \in M$, then $[\delta, \operatorname{ad} x] = 0$ for $M, M^{\perp}$ are ideals . But we saw above that

$$[\delta, \operatorname{ad} x] = \operatorname{ad}(\delta x)$$

so, for all $x \in L$, we have $\delta(x) = 0$; in other words, $\delta = 0$. □

7.5 Abstract Jordan Decomposition

Proposition 7.5.1

Let L be a complex Lie algebra. Suppose that δ is a derivation of L with Jordan decomposition $\delta = \sigma + \nu$, where σ is diagonalisable and ν is nilpotent. Then σ and ν are also derivations of L.

Proof. For $\lambda \in \mathbb{C}$, let

$$L_\lambda = \{x \in L : (\delta - \lambda)\}$$

L decomposes as a direct sum of generalised eigenspaces

$$L = \oplus_\lambda L_\lambda$$

As σ acts diagonalisably, then λ-eigenspace of σ is L_λ . Take $x \in L_\lambda$, $y \in L_\mu$, then

$$\begin{aligned}(\delta - (\lambda + \mu) 1_L)[x, y] &= [\delta x, y] + [x, \delta y] - [\lambda x, y] - [x, \mu y] \\ &= [(\delta - \lambda 1_L) x, y] + [x, (\delta - \mu 1_L y)]\end{aligned}$$

$(\delta - \lambda 1_L)^{m+n} [x, y]$ is the linear combination of the form $[(\delta - \lambda 1_L)^i x, (\delta - \mu 1_L y)^j y]$, which is zero. Thus $[x, y] \in L_{\lambda+\mu}$, so

$$\sigma([x, y]) = (\lambda + \mu)[x, y]$$

which is the same as

$$[\sigma(x), y] + [x, \sigma(y)] = [\lambda x, y] + [x, \mu y]$$

Thus σ is a derivation, and so is $\delta = \sigma = \nu$. □

Theorem 7.5.2

Let L be a complex semisimple Lie algebra. Each $x \in L$ can be written uniquely as $x = d + n$, where $d, n \in L$ are such that $\operatorname{ad} d$ is diagonalisable, $\operatorname{ad} n$ is nilpotent, and $[d, n] = 0$. Furthermore, if $y \in L$ commutes with x, then $[d, y] = 0$ and $[n, y] = 0$.

Proof. Let $\operatorname{ad} x = \sigma + \nu$, where $\sigma \in \mathfrak{gl}(L)$ is diagonalisable, $\nu \in \mathfrak{gl}(L)$ is niplpotent, and $[\sigma, \nu] = 0$. We know that σ and ν are derivations of the semisimple Lie algebra L. And we know that $\operatorname{ad} L = \operatorname{Der} L$, so there exsists $d, n \in L$ such that $\operatorname{ad} d = \sigma$ and $\operatorname{ad} n = \nu$. As ad is injective and

$$\operatorname{ad} x = \sigma + \nu = \operatorname{ad}(d + n)$$

we get that $x = d + n$.Moreover, $\operatorname{ad}[d, n] = [\operatorname{ad} d, \operatorname{ad} n] = 0$. The uniqueness of d and n follows from the uniqueness of the Jordan decomposition of $\operatorname{ad} x$. Suppose that $y \in L$ and that $(\operatorname{ad} x)\, y = 0$. Since σ and ν may be expressed as polynomials in $\operatorname{ad} x$. Let

$$\nu = c_0 1_L + c_1 \operatorname{ad} x + \cdots + c_r (\operatorname{ad} x)^r$$

Applying ν to y we see that $\nu(y) = c_0 y$. But ν is nilpotent and $\nu(x) = c_0 x$, so $c_0 = x$. Thus $\nu(y) = 0$, and so $\sigma(y) = (\operatorname{ad} x - \nu)\, y = 0$ also. $\square$

Definition 7.5.3

Let L be a semisimple Lie algebra and let $\theta : L \to \mathrm{gl}(V)$ be a representation of L. Suppose that $x \in L$ has abstract Jordan decomposition $x = d + n$. Then the Jordan decomposition of $\theta(x) \in \mathrm{gl}(V)$ is $\theta(x) = \theta(d) + \theta(n)$.

8 The Root Space Decomposition

8.1 Preliminary Results

Proposition 8.1.1

Supposet that L is a complex semisimple Lie algebra containing an abelian subalgebra H consisting of semisiple elements.
For each $\alpha \in H^*$, let

$$L_\alpha := \{x \in L : [h, x] = \alpha(h)x, \quad \forall h \in H\}$$

Let Φ denote the set of non-zero $\alpha \in L^*$ for which L_α is non-zero. We can write the decomposition of L into weight spaces for H as

$$L = L_0 \oplus \bigoplus_{\alpha \in \Phi} L_\alpha. \tag{$\star$}$$

Since L is finite-dimensional, this implies that Φ is finite.

Lemma 8.1.2

Suppose $\alpha, \beta \in H^*$, then

1. $[L_\alpha, L_\beta] = L_{\alpha+\beta}$
2. If $\alpha + \beta \neq 0$, then $\kappa(L_\alpha, L_\beta) = 0$.
3. The restrition of κ to L_0 is non-degenerate.

Proof. 1. Take $x \in L_\alpha, y \in L_\beta$, then

$$\begin{aligned}[h, [x, y]] &= -[y, [h, x]] - [x, [y, h]] \\ &= -[y, \alpha x] - [x, -\beta y] \\ &= (\alpha + \beta)[x, y]\end{aligned}$$

2. If $\alpha + \beta \neq 0$, then $\alpha + \beta \neq 0$. Take $x \in L_\alpha, y \in L_\beta$

$$\begin{aligned}\alpha(h)\kappa(x,y) &= \kappa(\alpha(h)x,y)\\ &= \kappa([h,x],y)\\ &= -\kappa([x,h],y)\\ &= -\kappa(x,[h.y])\\ &= -\kappa(x,\beta(h)y)\\ &= -\beta(h)\kappa(x,y)\end{aligned}$$

Thus

$$(\alpha(h) + \beta(h))\kappa(x,y) = 0$$

Then it follows that $\kappa(x,y) = 0$ since $\alpha(h) + \beta(h) \neq 0$

3. Suppose that $z \in L_0$, $\kappa(z, x_0) = 0$ for all $x_0 \in L_0$. From 2.,for each $\alpha \neq 0$, we have $\kappa(z, x_\alpha) = 0$ for all $x_\alpha \in L_\alpha$. Then for each element of x, we write

$$x = x_0 + \sum_{\alpha\in\Phi} x_\alpha$$

there is

$$\kappa(z,x) = \kappa(z,x_0) + \sum_{\alpha\in\Phi}\kappa(z,x_\alpha) = 0$$

Since L is semisimple, κ is non-degenerate(Cartan second Cretria). It follows that $z = 0$.

□

Exercise 8.1.3

Show that if $x \in L_\alpha$, where $\alpha \neq 0$, then ad_x is nilpotent.

Proof. Suppose that $\Phi = \{\alpha_0 := \alpha, \alpha_1, \cdots, \alpha_k\}$, then for each $j = 0, 1, \cdots, k$, since $\alpha_j \neq 0$, there exists $m_j \in \mathbb{Z}_{>0}$ such that $\alpha_j + m_j\alpha$ is different from any element of $\Phi \cup \{0\}$. Then

$$(\mathrm{ad}_x)^{m_j}\left(x_{\alpha_j}\right) = 0, \quad \forall j = 0,1,\cdots,k, \quad \forall x_{\alpha_j} \in L_{\alpha_j}$$

Further more,

$$(\mathrm{ad}_x)^{m_0+1}(x_0) = (\mathrm{ad}_x)^{m_0}([x,x_0]) = 0, \quad \forall x_0 \in L_0$$

since $[x, x_0] \in L_\alpha$. Set $N = \max\{m_0 + 1, m_1, \cdots, m_k\}$. Then

$$(\mathrm{ad}_x)^N (L) = (\mathrm{ad}_x)^N \left(L_0 + \oplus_{\alpha_j \in \Phi} L_{\alpha_j}\right) = 0$$

ad_x is nilpotent . □

Remark.
If H is small, then the decomposition $(\star)$ is likely to be rather coarse, with few non-zero weight spaces other than L_0. Furthermore, if H is properly contained in L_0, then we get little information about how the elements in L_0 that are not in H act on L. This is illustrated by the following exercise.

Exercise 8.1.4

Let $L = \mathrm{sl}(n, \mathbb{C})$, where $n \geq 2$, and let $H = \mathrm{Span}\{h\}$, where $h = e_{11} - e_{22}$. Find $L_0 = C_L(H)$, and determine the direct sum decomposition $(\star)$ with respect to H.

Solution. Take $x \in \mathfrak{sl}(n, \mathbb{C})$, suppose $x = \sum_{i \neq j, \text{and } i,j \neq 1,2} x^{ij} x_{ij} + x^{12} e_{12} + x^{21} e_{21} + y^k \sum_{k=2}^{n} (e_{11} - e_{kk})$. For each $1 \leq i \neq j \leq n$ If $i, j \neq 1, 2$, then

$$[e_{11}, e_{ij}] = e_{11} e_{ij} - e_{ij} e_{11} = 0$$

$$[e_{22}, e_{ij}] = e_{22} e_{ij} - e_{22} e_{ij} = 0$$

$$[h, x^{12} e_{12}] = x^{12} [h, e_{12}] = x^{12} (e_{11} e_{12} - e_{12} e_{11} - e_{22} e_{12} + e_{12} e_{22}) = 2x^{12} e_{12}$$

$$[h, x^{21} e_{21}] = x^{21} [h, e_{21}] = x^{21} (e_{11} e_{21} - e_{21} e_{11} - e_{22} e_{21} + e_{21} e_{22}) = -2x^{21} e_{21}$$

$$[h, e_{11} - e_{kk}] = [e_{22}, e_{kk}] = 0$$

Then $x \in L_0 \iff x^{12} = x^{21}$.

$L_0 = \mathrm{span}\left(\{e_{ij}, i \neq j, \text{and } i, j \neq 1, 2\} \cup \{e_{12} - e_{21}\} \cup \{e_{11} - e_{kk} : k = 2, 3, \cdots, n\}\right)$

◊

8.2 Cartan Subalgebras

Definition 8.2.1

A Lie subalgebra H of a Lie algebra L is said to be a ***Cartan subalgebra*** (or CSA) if H is abelian and every element $h \in H$ is semisimple, and moreover H is maximal with these properties.

Lemma 8.2.2

Let H be a Cartan subalgebra of L. Suppose that $h \in H$ is such that the dimension of $C_L(h)$ is minimal. Then every $s \in H$ is central in $C_L(h)$, and so $C_L(h) \subseteq C_L(s)$. Hence $C_L(h) = C_L(H)$.

Theorem 8.2.3

If H is a Cartan subalgebra of L and $h \in H$ is such that $C_L(h) = C_L(H)$, then $C_L(h) = H$. Hence H is self-centralising.

8.3 Definition of the Root Space Decomposition

Definition 8.3.1

Let H be a Cartan subalgebra of our semisimple Lie algebra L. As $H = C_L(H)$, the direct sum decomposition of L into weight spaces for H considered in §10.1 may be written as

$$L = H \oplus \bigoplus_{\alpha \in \Phi} L_\alpha,$$

where Φ is the set of $\alpha \in H^*$ such that $\alpha \neq 0$ and $L_\alpha \neq 0$. Since L is finite-dimensional, Φ is finite.

- If $\alpha \in \Phi$, then we say that α is a ***root*** of L and L_α is the associated ***root space***.
- The direct sum decomposition above is the ***root space decomposition***.

Remark.
It should be noted that the roots and root spaces depend on the choice of Cartan subalgebra H.

Lemma 8.3.2

Let $x, y : V \to V$ be a linear maps from a complex vector space V to itself. Suppose that x and y both commute with $[x, y]$. Then $[x, y]$ is a nilpotent map.

Proof. Let $S = \operatorname{span}\{x, y\}$, then

$$S' = \operatorname{span}\{[x, y]\}$$

commuts with all element of S, that is $[S', S] = 0$, it follows that S is nilpotent, thus it is solvable. From Lie's theorem, there is a basis of V in which x, y is represented upper triangular. Let $L_i = \operatorname{span}\{e_1, \cdots, e_i\}$, then L_i is x-invariant and y-invariant, $i = 1, \cdots, n$. There is x^{ii} , y_{ii}, $i = 1, \cdots, n$ such that

$$x(e_i) + L_{i-1} = x^{ii}e_i, \quad y(e_i) + L_{i-1} = y^{ii}e_i.$$

Then

$$[x, y](e_i) + L_{i-1} = x^{ii}y^{ii}e_i - y^{ii}x^{ii}e_i + L_{i-1} = 0 + L_{i-1}$$

Thus $[x, y](e_i) \in L_{i-1}$. Thus $[x, y]$ is nilpotent. $\square$

Lemma 8.3.3

Suppose that $\alpha \in \Phi$ and that x is a non-zero element in L_α. Then $-\alpha$ is a root and there exists $y \in L_{-\alpha}$ such that Span$\{x, y, [x, y]\}$ is a Lie subalgebra of L isomorphic to sl$(2, \mathbb{C})$.

Proof. κ is non-degenerate, there exists $w = y_0 + \sum_{\beta\in\Phi} y_\beta$ such that $\kappa(x, w) \neq 0$. But for each $\beta \neq \alpha$, $\kappa\left(x, y_\beta\right) = 0$, there must be $-\alpha \in \Phi$, $\kappa(x, y_{-\alpha}) = \kappa(x, w) \neq 0$. We take $y = y_{-\alpha}$. Since $\alpha \neq 0$, there exists $t \in H$ such that $\alpha(t) \neq 0$. Then

$$\kappa(t, [x, y]) = \kappa([t, x], y) = \alpha(t)\kappa(x, y) \neq 0$$

thus $[x, y] \neq 0$.

Let $S := \operatorname{span}\{x, y, [x, y]\}$, $[x, y]$ lies in $L_0 = H$. As x and y are simultaneuous eigenvectors for all elements of ad H, and so in particular for ad$[x, y]$, this shows that S is a Lie subalgebra of L. It remains to show that S is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

Let $h := [x, y] \in S$. We claim that $\alpha(h) \neq 0$. If not , then $[h, x] = \alpha(h)x = 0$;

similarly $[x, y] = -\alpha(h)x = 0$, so $\operatorname{ad} h : L \to L$ commutes with $\operatorname{ad} x$ and $\operatorname{ad} y$, thus is nilpotent. The other hand, because H is a Cartan subslagebra, h is semisimple . But the only element of L that is both semisimple and nilpotent is 0, so $h = 0$, a contradiction.

Thus S is a 3-dimensional complex Lie algebra with $S' = S$. S is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. □

Exercise 8.3.4

Show that for each $\alpha \in \Phi$, S_α has a basis $\{e_\alpha, f_\alpha, h_\alpha\}$ such that

1. $e_\alpha \in L_\alpha, f_\alpha \in L_{-\alpha}, h_\alpha \in H$ and $\alpha(h_\alpha) = 2$.
2. The map $\theta : S_\alpha \to \mathfrak{sl}(2, \mathbb{C})$, defined by $\theta(e_\alpha) = e$, $\theta(f_\alpha) = f$, $\theta(h_\alpha) = h$ is a Lie algebra isomorphism.

Proof. Let $x \in L_\alpha$. Since κ is non-degenerate, there exists $w = y_0 + \sum_{\alpha\in\Phi} y_\alpha$ such that $\kappa(x, w) \neq 0$, then

$$\kappa(x, w) = \kappa\left(x, y_0 + \sum_{\alpha\in\Phi} y_\alpha\right) = \kappa(x, y_{-\alpha}) \neq 0$$

Denote $y = y_{-\alpha}$.

$[x, y] \in L_{\alpha+(-\alpha)} = L_0 = H$. Take $h = [x, y]$. $h(x) = \alpha(h)x$

$$\kappa(h(x), y) = \alpha(h)\kappa(x, y) \neq 0$$

Then $\alpha(h) \neq 0$. Take $e_\alpha = x$, $f_\alpha = \frac{2}{\alpha(h)}y$. $h_\alpha = [e_\alpha, f_\alpha]$. Then

$$h_\alpha(e_\alpha) = \frac{2}{\alpha(h)}h(x) = 2x = 2e_\alpha$$

Let $S_\alpha = \operatorname{span}\{e_\alpha, f_\alpha, h_\alpha\}$. Then $S'_\alpha = S_\alpha$. S is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

$$[e_\alpha, f_\alpha] = h_\alpha, \quad [h_\alpha, e_\alpha] = 2e_\alpha, \quad [h_\alpha, f_\alpha] = 2f_\alpha$$

□

8.4 Root Strings and Eigenvalues

Lemma 8.4.1

Given $h \in H$, let θ_h denote the map $\theta_h \in H^*$ defined by

$$\theta_h(k) = \kappa(h, k), \quad \forall k \in H$$

Then it is a isomorphism since $\kappa|_H$ is non-degenerate. In particular , for $\alpha \in H^*$, there is a unique element $t_\alpha \in H$ such that

$$\kappa(t_\alpha, k) = \alpha(k), \quad \forall k \in H$$

Lemma 8.4.2

Let $\alpha \in \Phi$. If $x \in L_\alpha$, $y \in L_{-\alpha}$, then $[x, y] = \kappa(x, y)\, t_\alpha$. In particular $h_\alpha = [e_\alpha, f_\alpha] \in \operatorname{span}\{t_\alpha\}$.

Proof. For $h \in H$, we have

$$\kappa(h, [x, y]) = \alpha(h)\kappa(x, y) = \kappa(t_\alpha, h)\kappa(x, y) = \kappa(h, \kappa(x, y)\, t_\alpha)$$

This shows that $[x, y] - \kappa(x, y)\, t_\alpha \in H^\perp = 0$. Thus $[x, y] = \kappa(x, y)\, t_\alpha$ □

Definition 8.4.3

Let α be a root. We may regard L as an $\mathfrak{sl}(\alpha)$-modul via restriction of the adjoint representation. Thus if $\alpha \in \mathfrak{sl}(\alpha)$ and $y \in L$, then the action is given by

$$a \cdot y = (\operatorname{ad} a)(y) = [a, y]$$

Remark.

- The $\mathfrak{sl}(\alpha)$-submodules of L are precisely the vector subspaces M of L such that $[s, m] \in M$ for all $s \in \mathfrak{sl}(\alpha)$ and $m \in M$.

Lemma 8.4.4

If M is a $\mathfrak{sl}(\alpha)$-submodule of L, then the eigenvalues of h_α acting on M are intergers.

Proof. Since M is finite-dimensional, by Wely's Theorem, M may be decomposed into a direct sum of irreducible $\mathfrak{sl}(\alpha)$ -modules; for irreducible $\mathfrak{sl}(2, \mathbb{C})$-

modules, the eigenvalues of h acting on M are integers. The the lemma follows from the isomorphism between $\mathfrak{sl}(\alpha)$ and $\mathfrak{sl}(2,\mathbb{C})$. □

Definition 8.4.5

If $\beta \in \Phi$ or $\beta = 0$, let

$$M := \oplus_c L_{\beta+c\alpha}$$

where the sum is over all $c \in \mathbb{C}$ such that $\beta + c\alpha \in \Phi$. M is an $\mathfrak{sl}(\alpha)$-submodule of L. THis module is said to be the ***α-root string through mβ***
.

Proof. For each $c_1, c_2 \in \mathbb{C}$. If $\beta \neq 0$, then $\left[L_{\beta+c_1\alpha}, L_{\beta+c_2\alpha}\right] \in L_{\beta+(\beta^{-1}\alpha+c_1+c_2)\alpha}$
. If $\beta = 0$, then $\left[L_{\beta+c_1\alpha}, L_{\beta+c_2\alpha}\right] \in L_{\beta+(c_1+c_2)\alpha}$ □

Proposition 8.4.6

Let $\alpha \in \Phi$. The root spaces $L_{\pm\alpha}$ are 1-dimensional. Moreover, the only multiples of α which lie in Φ are $\pm\alpha$.

9

Low Dimension Lie Algebras

9.1 Dimension 3

9.1.1 Lie Algebras where $L' = L$

Theorem 9.1.1

Suppose that L is a complex Lie algebra of dimension 3 such that $L = L'$. Then L is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

Proof.

1. Let $x \in L$ be non-zero. We claim that $\operatorname{ad} x$ has rank 2. Extend x to a basis of L say $\{x, y, z\}$. Then L' is spanned by

$$\{[x, y], [x, z], [y, z]\}$$

But $L' = L$, this set must be linearly independent, and hence the imaeg of $\operatorname{ad} x$ has a basis $\{[x, y], [x, z]\}$ of size 2, as required.

2. We claim that there is some $h \in L$ such that $\operatorname{ad} h$ has an eigenvector with a non-zero eiganvalue. Choose any non-zero $x \in L$. If $\operatorname{ad} x$ has a non-zero eigenvalue, then we take $h = x$. If $\operatorname{ad} x$ has no non-zero eigenvalues, then, as it has rank 2, its Jordan canonical form must be

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

THis martix indicates there is a basis of L extending $\{x\}$, say $\{x, y, z\}$ such that $[x, y] = x$ and $[x, z] = y$. So $\operatorname{ad} y$ has x as an eigenvector with eigenvalue -1, and we may take $h = y$.

3. By the previous step, we may find $h, x \in L$ such that $[h, x] = \alpha x \neq 0$. Since $h \in L$ and $L = L'$, we know $\operatorname{ad} h$ has trace zero. It follows that $\operatorname{ad} h$ must have three distinc eigenvalues $\alpha, 0, -\alpha$. If y is an eigenvector for $\operatorname{ad} h$ with eigenvalue $-\alpha$, then $\{h, x, y\}$ is a basis of L. In this basis, $\operatorname{ad} h$ is represented by a diagonal matrix.

4. To determine $[x, y]$. Note that

$$[h, [x, y]] = [[h, x], y] + [x, [h, y]] = \alpha[x, y] + (-\alpha)[x, y] = 0$$

5. We now make two applications of step 1. Firstly, $\ker \operatorname{ad} h = \operatorname{span}\{h\}$, so $[x, y] = \lambda h$. Seconedly, $\lambda \neq 0$ other wise $\ker \operatorname{ad} x$ is 2-dimensional. By replacing x with $\lambda^{-1}x$ we may assume that $\lambda = 1$.

□

Proposition 9.1.2

Suppose that $\alpha, \beta \in \Phi$ and $\beta \neq \pm\alpha$.

1. $\beta(h_\alpha) \in \mathbf{Z}$.
2. There are integers $r, q \geq 0$ such that $\beta + k\alpha \in \Phi$ if and only if $k \in \mathbf{Z}$ and $-r \leq k \leq q$. Moreover, $r - q = \beta(h_\alpha)$.
3. If $\alpha + \beta \in \Phi$, then $[e_\alpha, e_\beta]$ is a non-zero scalar multiple of $e_{\alpha+\beta}$.
4. $\beta - \beta(h_\alpha)\alpha \in \Phi$.

Proof. Let $M := \bigoplus_k L_{\beta+k\alpha}$ be the root string of α through β.

1. $\beta(h_\alpha)$ is the eigenvalue of h_α acting on L_β , so it lies in $\mathbb{Z}$.
2. $\dim L_{\beta+k\alpha} = 1$ whenever $\beta + k\alpha$ is a root, so the eigenspaces of $\operatorname{ad} h_\alpha$ on M are all 1-dimensional and , since $(\beta + k\alpha) h_\alpha = \beta(h_\alpha) + 2k$, the eigenvalues of $\operatorname{ad} h_\alpha$ on M are either all even or all odd.

□

9.2 Cartan Subalgebras as Inner-Product Spaces

10 Root Systems